

Section 2.2. Axioms of Probability

Definition (Axiom of Probability)

Axiom 1 Nonnegativity: For any event A , $P(A) \geq 0$

Axiom 2 Normalization: $P(S) = 1$

Axiom 3 Countable Additivity: If $A_1, A_2, A_3, \dots$ is an infinite collection of disjoint events, then

$$P\left(\bigcup_{i=1}^{\infty} A_i\right) = \sum_{i=1}^{\infty} P(A_i) \quad (1)$$

Note: Direct consequences the above axioms are below Propositions. You do not need to learn the proof of any of the following propositions, but you should know all the propositions and we will use them to calculate the probabilities.

Proposition 1

$$P(\phi) = 0$$

proof. Consider the sequence $\{A_i\}_{i \in \mathbb{N}}$,
 $A_i = \emptyset$. Now $A_i \cap A_j = \emptyset$. By Axiom 3

$$p(\emptyset) = \sum_{i=1}^{\infty} p(\emptyset)$$
$$0 = \sum_{i=2}^{\infty} p(\emptyset)$$

By axiom 1 $p(A) \geq 0$, for every $A \in \mathcal{S}$,
this is only possible if $p(\emptyset) = 0$

Proposition 2

If $A_1, A_2, A_3, \dots, A_n$ is a finite collection of disjoint events, then

$$P\left(\bigcup_{i=1}^n A_i\right) = \sum_{i=1}^n P(A_i)$$

Let $A_i = \emptyset$ for $i \geq n+1$. Now by axiom 3

$$P\left(\bigcup_{i=1}^{\infty} A_i\right) = \sum_{i=1}^{\infty} P(A_i)$$

Since $A_i = \emptyset$ for $i \geq n+1$, $\bigcup_{i=1}^{\infty} A_i = \bigcup_{i=1}^n A_i$

$P(A_i) = 0$ for $i \geq n+1$, hence

$$P\left(\bigcup_{i=1}^{\infty} A_i\right) = \sum_{i=1}^n P(A_i)$$

Proposition 3

$$P(A') = 1 - P(A)$$

$$A' \cup A = S \quad \text{and} \quad A \cap A' = \emptyset$$

By proposition 2

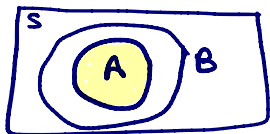
$$P(A' \cup A) = P(A') + P(A)$$
$$P(S) = P(A') + P(A)$$

$$1 = P(A') + P(A)$$

$$\Rightarrow P(A') = 1 - P(A)$$

Proposition 4

If $A \subset B$, then $P(A) \leq P(B)$



$$A \subset B \Rightarrow B = A \cup (B-A) \quad \text{and} \quad (B-A) \cap A = \emptyset$$

$$\Rightarrow P(B) = P(A) + P(B-A)$$

$$\text{we know } P(B-A) \geq 0 \Rightarrow P(A) \leq P(B)$$

Proposition 5

For any event A , $P(A) \leq 1$.

Any given event $A \subseteq S$,

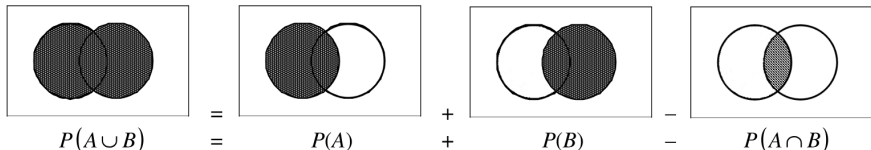
$$P(A) \leq P(S) \text{ and } P(S)=1$$

$\Rightarrow$

$$P(A) \leq 1$$

Proposition 6

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$



Proposition 7

$$\begin{aligned} P(A \cup B \cup C) = & P(A) + P(B) + P(C) \\ & - P(A \cap B) - P(A \cap C) - P(B \cap C) \\ & + P(A \cap B \cap C) \end{aligned}$$

Definition (Uniform probability Law (discrete case))

If a sample space S is finite with $|S|$ possible outcomes and all the outcomes are equally likely, then probability of event E can be calculated from

$$P(E) = \frac{|E|}{|S|}$$

$|E|$ means number of possible outcomes that E contains.

Eg 1. You toss a standard six-sided die. The sample space is $S = \{1, 2, 3, 4, 5, 6\}$, 6 simple events that are equally likely.

- (a) A = the number rolled is even. Calculate $P(A)$.
- (b) B = the number rolled is at least 2. Calculate $P(B)$.
- (c) C = the number rolled is more than 3. Calculate $P(C)$.

$$|S| = 6$$

$$A = \{2, 4, 6\} \quad P(A) = \frac{|A|}{|S|} = \frac{3}{6} = \frac{1}{2}$$

$$B = \{2, 3, 4, 5, 6\} \quad P(B) = \frac{|B|}{|S|} = \frac{5}{6}$$

$$C = \{4, 5, 6\} \quad P(C) = \frac{|C|}{|S|} = \frac{3}{6} = \frac{1}{2}$$

Definition (Uniform probability Law(continuous form))

Sample space is uncountable and infinite with $\text{Area}(S)$ and all the outcomes are equally likely. For event E we have

$$P(E) = \frac{\text{Area}(E)}{\text{Area}(S)}$$

Eg 2: $S = \{(x, y) : 0 \leq x, y \leq 1\}$. Find

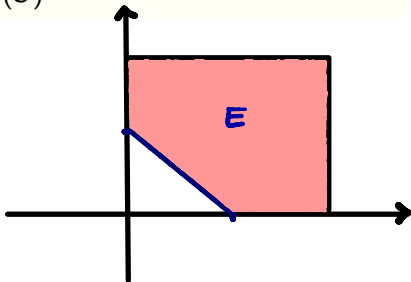
(a) $P(\{(x, y) : x + y \geq \frac{1}{2}\})$

(b) $P(\{(0.5, 0.3)\})$

$$E = \{(x, y) : x + y \geq \frac{1}{2}\}$$

E is the highlighted part.

It's hard to calculate probability of E directly.



since it's hard to calculate the area of E ,
so we will use the formula of

$$P(E) = 1 - P(E^c)$$

E^c is the triangle 

$$\text{Area}(E^c) = \frac{\frac{1}{2} \cdot \frac{1}{2}}{2} = \frac{1}{8}$$

$$P(E^c) = \frac{\text{Area}(E^c)}{\text{Area}(S)} = \frac{\frac{1}{8}}{1} = \frac{1}{8}$$

$$\Rightarrow P(E) = 1 - \frac{1}{8} = \frac{7}{8}.$$

(b) Area of a single point is zero. $P(\{(0.5, 0.3)\}) = 0$

Eg 3. Consider the experiment of two rolls of a tetrahedral die. Let every possible outcome have probability $\frac{1}{16}$. If we show the outcome of the first die with X and outcome of the second die with Y . Calculate the following probabilities

(a) $P(X = 1)$

(b) Let $Z = \max(X, Y)$, find $P(Z = 3)$.

$$S = \{ (1,1), (1,2), (1,3), (1,4), \\ (2,1), (2,2), (2,3), (2,4), \\ (3,1), (3,2), (3,3), (3,4), \\ (4,1), (4,2), (4,3), (4,4) \} \quad |S| = 16.$$

$$\text{Event}(X=1) = A = \{ (1,1), (1,2), (1,3), (1,4) \} \quad P(A) = \frac{4}{16}$$

$$\text{Event}(\max(X, Y) = 3) = B = \{ (3,1), (3,2), (3,3), (1,3), (2,3) \}$$

$$P(B) = \frac{5}{16}$$

Eg 4. Consider the Sample space $S = \{1, 2, \dots\}$. We are given $P(n) = \frac{1}{2^n}$.
Calculate $P(\text{outcome is even}) =$

Recall Geometric series :

$$\sum_{k=m}^{\infty} r^k = \frac{r^m}{1-r} ; \text{ if } |r| < 1$$

Event=outcome is even = $A = \{2, 4, 6, 8, 10, \dots\}$

we can think of A as a countable union of disjoint sets,

meaning $A = \{2\} \cup \{4\} \cup \{6\} \cup \{8\} \dots$

so $P(A) = P(\{2\}) + P(\{4\}) + P(\{6\}) + P(\{8\}) + \dots$

$$P(A) = \frac{1}{2^2} + \frac{1}{2^4} + \frac{1}{2^6} + \frac{1}{2^8} + \dots$$

$$P(A) = \frac{1}{2^2} \left(1 + \frac{1}{2^2} + \frac{1}{2^4} + \frac{1}{2^6} + \dots \right)$$

$$P(A) = \frac{1}{4} \left(\sum_{k=0}^{\infty} \left(\frac{1}{4}\right)^k \right)$$

$$P(A) = \frac{1}{4} \cdot \frac{1}{1 - \frac{1}{4}} = \frac{1}{4} \cdot \frac{1}{\frac{3}{4}} = \frac{1}{4} \cdot \frac{4}{3} = \frac{1}{3}$$

Eg 5: You pick a card from a standard deck of 52 cards.

(4 suits: spades (S), hearts (H), diamonds (D), clubs (C);

13 cards in each suit: ace (A), king (K), queen (Q), jack (J), 10, 9, 8, 7, 6, 5, 4, 3, 2)

$S = \{A - S, A - H, A - D, A - C, K - S, K - H, K - D, K - C, Q - S, Q - H, Q - D, Q - C, \dots, 2 - S, 2 - H, 2 - D, 2 - C\}$.

$N(S) = 52$.

The 52 outcomes are equally likely. Calculate probabilities for each described event.

(a) A = the card is an Ace

$$P(A) = \frac{|A|}{|S|} = \frac{4}{52}$$

(b) B = the card is a Spade

$$P(B) = \frac{13}{52}$$

(c) C = the card is the Ace of Spades

$$P(C) = \frac{1}{52}$$

(d) D = the card is an Ace or a Spade

note that $C = A \cap B$.

$$D = A \cup B$$

$$D = A \cup B$$

$$P(D) = P(A) + P(B) - P(A \cap B)$$

$$P(D) = \frac{4}{52} + \frac{13}{52} - \frac{1}{52} = \frac{16}{52} = \frac{4}{13}$$